

EX NO:	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

```
clc; clear;
```

```
Wp=0.4; Ws=0.55; Rp=1; As=40; EX NO:2
```

```
Chebyshev IIR Filter
```

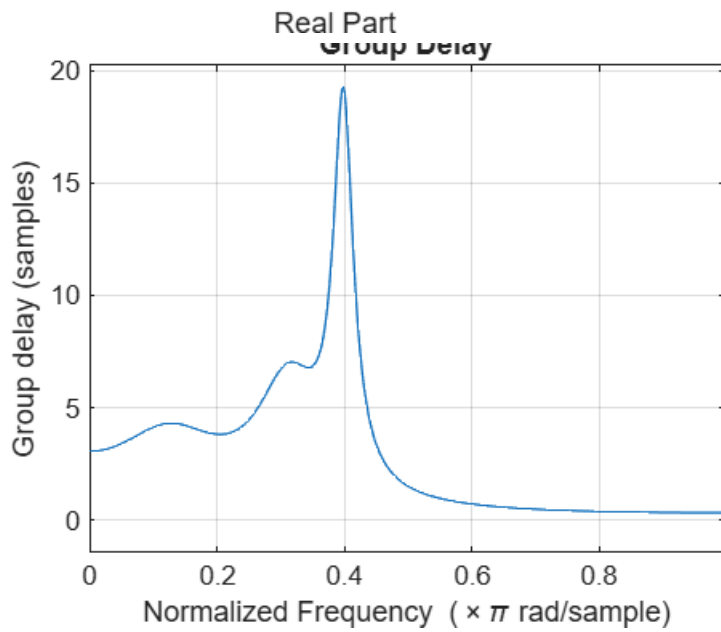
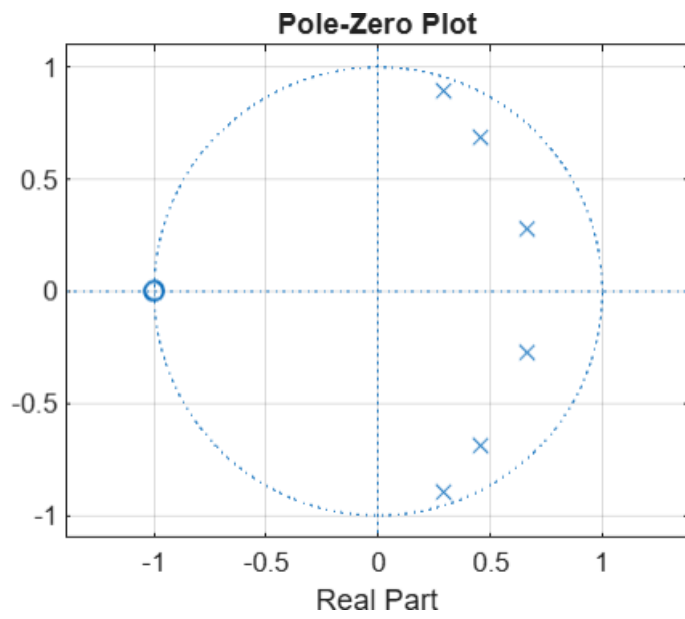
```
DATE: 2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass
```

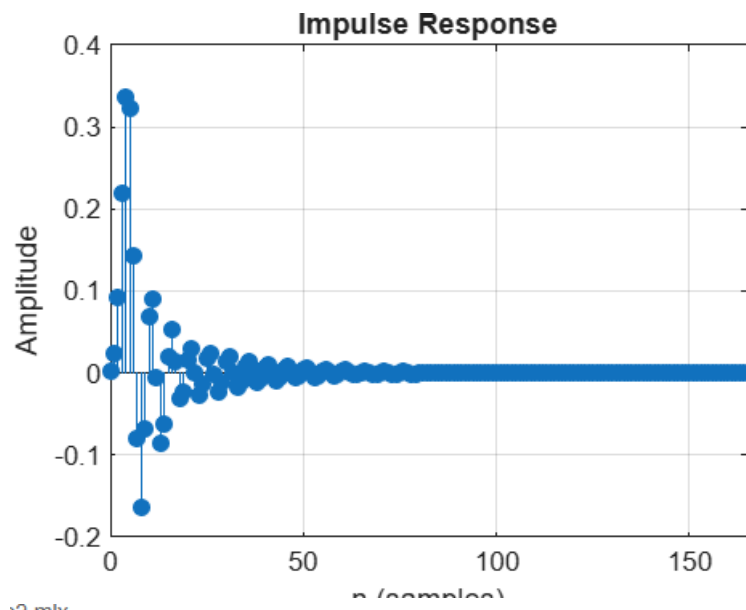
```
Filter using MATLAB
```

```
[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);
```

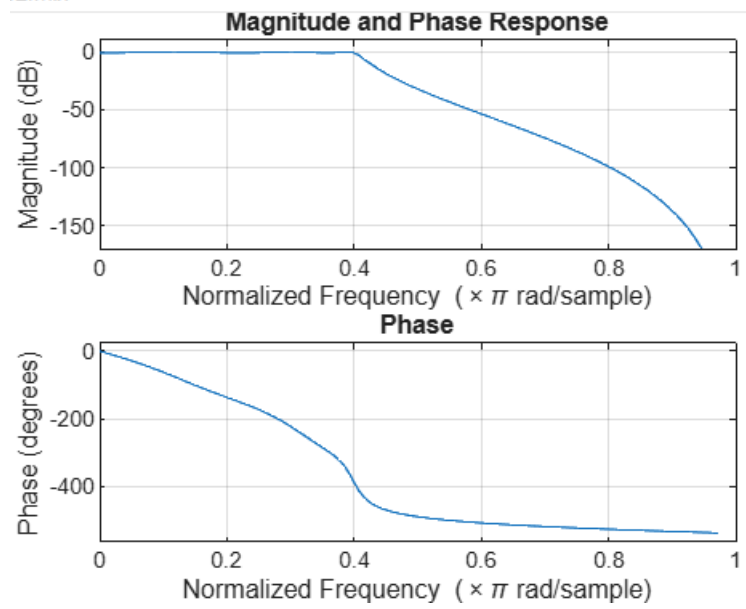
```
fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
```

```
if all(abs(roots(a))<1) disp('System is STABLE'); else
    disp('System is UNSTABLE'); end
```





y2.mlx



Ex no 2	Chebyshev IIR Filter
date	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

```
clc; clear; close all;
```

```
% Specifications
```

```
Rp = 1; % Passband Ripple (dB)
```

```
Rs = 40; % Stopband Attenuation (dB) Wp = 0.5; % Passband Edge
```

```
EX NO:2 Chebyshev IIR Filter
```

```
DATE: 2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR  
High-Pass
```

```
Filter
```

```
Ws = 0.35; % Stopband Edge
```

```
% Find Filter Order
```

```
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
```

```
% Design High-Pass Filter [b,a] = cheby1(n,Rp,Wn,'high');
```

```
% Frequency Response figure;
```

```
freqz(b,a);
```

```
title('Frequency Response');
```

```
% Impulse Response figure;
```

```
impz(b,a);
```

```
title('Impulse Response');
```

```
% Group Delay figure;
```

```
grpdelay(b,a); title('Group Delay');
```

```
% Pole-Zero Plot figure; zplane(b,a);
```

```
title('Pole-Zero Plot');
```

```
fprintf('Filter Order = %d\n',n);
```

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

